PES UNIVERSITY

CS-AIML

OS ORANGE PROBLEM 2

PES-VERSION-CONTROL-SYSTEM

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1.Objective

The objective of this project is to design and implement a simplified version of a Git-like Version Control System (VCS).
This system tracks file changes, stores snapshots efficiently, and maintains commit history using core operating system and filesystem concepts

2. Introduction

Version control systems like Git help in tracking changes in files and maintaining different versions of a project.
This project implements a mini version of Git called **PES-VCS**, which uses content-addressable storage and filesystem structures to manage data.Instead of storing differences, the system stores **complete snapshots**, making retrieval and history tracking efficient.

3. Phase Implementation

Phase 1: Object Storage Foundation

1A

```
thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem$ export PES_AUTHOR="PES1UG24AM476"
thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem$ echo $PES_AUTHOR
PES1UG24AM476
thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem$ make clean
rm -rf .pes
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objects.o test_tree.o
rm -rf .pes
thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem$ make test_objects
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -o test_objects test_objects.o object.o
thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem$ ./test_objects
Stored blob with hash: 57abd5ea75baddee57abd5ea75baddee57abd5ea75baddee57abd5ea75baddee
Object stored at: .pes/objects/57/abd5ea75baddee57abd5ea75baddee57abd5ea75baddee57abd5ea75baddee
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem$
```

1B

```
thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem$ find .pes/objects -type f
.pes/objects/f3/f97cbe5faf57abf3f97cbe5faf57abf3f97cbe5faf57abf3f97cbe5faf57ab
.pes/objects/57/abd5ea75baddee57abd5ea75baddee57abd5ea75baddee57abd5ea75baddee
.pes/objects/ad/d66bb5da6d361badd66bb5da6d361badd66bb5da6d361badd66bb5da6d361b
thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem$
```

Phase 2: Tree Objects

2A

```
thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem$ make test_tree
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -o test_tree test_tree.o object.o tree.o
thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem$
```

2B

```
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem$ find .pes/objects -type f
.pes/objects/f3/f97cbe5faf57abf3f97cbe5faf57abf3f97cbe5faf57abf3f97cbe5faf57ab
.pes/objects/57/abd5ea75baddee57abd5ea75baddee57abd5ea75baddee57abd5ea75baddee
.pes/objects/ad/d66bb5da6d361badd66bb5da6d361badd66bb5da6d361badd66bb5da6d361b
thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem$ xxd .pes/objects/f3/f97cbe5faf57abf3f97cbe5faf57abf3f97cbe5faf57abf3f97cbe5faf57ab  | head -
20
00000000: 626c 6f62 2031 3500 5465 7374 2069 6e74  blob 15.Test int
00000010: 6567 7269 5879 0a                        egriXy.
thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem$
```

Phase 3: The Index (Staging Area)

3A

```
thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem$ mkdir test_pes
cd test_pes
thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem/test_pes$ cp ../pes .
thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem/test_pes$ ./pes init
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status
Initialized empty PES repository in .pes/
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)

thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem/test_pes$
```

3B

```
thrupthi-g-d@Lily-1: ~/PES1UG24AM476/os-u4-orange-problem/test_pes
~/PES1UG24AM476/os-u4-orange-problem/test_pes

Untracked files:
  (nothing to show)

thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem/test_pes$ cat .pes/index
100664 d8ec763b9d4e2793d8ec763b9d4e2793d8ec763b9d4e2793d8ec763b9d4e2793 1776435969 6 file1.txt
100664 2c964b259249a4d22c964b259249a4d22c964b259249a4d22c964b259249a4d2 1776435969 6 file2.txt
thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem/test_pes$ ./pes commit -m "Initial commit"
```

Phase 4: Commits and History

4A

```
thrupthi-g-d@Lily-1: ~/PES1UG24AM476/os-u4-orange-problem/test_pes
~/PES1UG24AM476/os-u4-orange-problem/test_pes

Untracked files:
  (nothing to show)

thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem/test_pes$ cat .pes/index
100664 d8ec763b9d4e2793d8ec763b9d4e2793d8ec763b9d4e2793d8ec763b9d4e2793 1776435969 6 file1.txt
100664 2c964b259249a4d22c964b259249a4d22c964b259249a4d22c964b259249a4d2 1776435969 6 file2.txt
thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem/test_pes$ ./pes commit -m "Initial commit"
echo "update" >> file1.txt
./pes add file1.txt
./pes commit -m "Update file1"
echo "bye" > file3.txt
./pes add file3.txt
./pes commit -m "Add file3"
./pes log
Committed: 924924924924... Initial commit
Committed: 31984ca6d3e9... Update file1
Committed: dbed763b1d8e... Add file3
commit dbed763b1d8e47a3dbed763b1d8e47a3dbed763b1d8e47a3dbed763b1d8e47a3
Author: PES1UG24AM476
Date:   1776436074

    Add file3

Add file3

commit 31984ca6d3e9f47a31984ca6d3e9f47a31984ca6d3e9f47a31984ca6d3e9f47a
Author: PES1UG24AM476
Date:   1776436074

    Update file1dd file3

commit 92492492492492c992492492492492c992492492492492c992492492492492c9
Author: PES1UG24AM476
Date:   1776436074

    Initial commit◆

thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem/test_pes$
```

4B

```
    Initial commit◆

thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem/test_pes$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/02/01000080402010020100008040201002010000804020100201000080402010
.pes/objects/05/02010080c060b00502010080c060b00502010080c060b00502010080c060b0
.pes/objects/2c/964b259249a4d22c964b259249a4d22c964b259249a4d22c964b259249a4d2
.pes/objects/31/984ca6d3e9f47a31984ca6d3e9f47a31984ca6d3e9f47a31984ca6d3e9f47a
.pes/objects/73/399c4e271389c473399c4e271389c473399c4e271389c473399c4e271389c4
.pes/objects/92/492492492492c992492492492492c992492492492492c992492492492492c9
.pes/objects/d8/ec763b9d4e2793d8ec763b9d4e2793d8ec763b9d4e2793d8ec763b9d4e2793
.pes/objects/db/ed763b1d8e47a3dbed763b1d8e47a3dbed763b1d8e47a3dbed763b1d8e47a3
.pes/refs/heads/main
thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem/test_pes$
```

4C

```
.pes/objects/92/492492492492c992492492492492c992492492492492c992492492492492c9
.pes/objects/d8/ec763b9d4e2793d8ec763b9d4e2793d8ec763b9d4e2793d8ec763b9d4e2793
.pes/objects/db/ed763b1d8e47a3dbed763b1d8e47a3dbed763b1d8e47a3dbed763b1d8e47a3
.pes/refs/heads/main
thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem/test_pes$ cat .pes/refs/heads/main
cat .pes/HEAD
dbed763b1d8e47a3dbed763b1d8e47a3dbed763b1d8e47a3dbed763b1d8e47a3
ref: refs/heads/main
thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem/test_pes$
```

FINAL INTEGRATION

thrupthi-g-d@Lily-1: ~/PES1UG24AM476/os-u4-orange-problem
~/PES1UG24AM476/os-u4-orange-problem

```
--- Second Commit ---
Committed: e773b95c2e17... Update file.txt

--- Third Commit ---
Committed: 62b158ac56ab... Add farewell

--- Full History ---
commit 62b158ac56ab55aa62b158ac56ab55aa62b158ac56ab55aa62b158ac56ab55aa
Author: PES1UG24AM476
Date:   1776436530

    Add farewelldd farewel!

commit e773b95c2e170b85e773b95c2e170b85e773b95c2e170b85e773b95c2e170b85
Author: PES1UG24AM476
Date:   1776436530

    Update file.txtfarewel

commit ec763b1d8ec763b1ec763b1d8ec763b1ec763b1d8ec763b1ec763b1d8ec763b1
Author: PES1UG24AM476
Date:   1776436530

    Initial commit◆


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
62b158ac56ab55aa62b158ac56ab55aa62b158ac56ab55aa62b158ac56ab55aa

--- Object Store ---
Objects created:
8
.pes/objects/05/02010080c060b00502010080c060b00502010080c060b00502010080c060b0
.pes/objects/62/b158ac56ab55aa62b158ac56ab55aa62b158ac56ab55aa62b158ac56ab55aa
.pes/objects/66/b3592c168bc5e266b3592c168bc5e266b3592c168bc5e266b3592c168bc5e2
```

thrupthi-g-d@Lily-1: ~/PES1UG24AM476/os-u4-orange-problem
~/PES1UG24AM476/os-u4-orange-problem

```
--- Second Commit ---
Committed: e773b95c2e17... Update file.txt

--- Third Commit ---
Committed: 62b158ac56ab... Add farewell

--- Full History ---
commit 62b158ac56ab55aa62b158ac56ab55aa62b158ac56ab55aa62b158ac56ab55aa
Author: PES1UG24AM476
Date:   1776436530

    Add farewelldd farewel!

commit e773b95c2e170b85e773b95c2e170b85e773b95c2e170b85e773b95c2e170b85
Author: PES1UG24AM476
Date:   1776436530

    Update file.txtfarewel

commit ec763b1d8ec763b1ec763b1d8ec763b1ec763b1d8ec763b1ec763b1d8ec763b1
Author: PES1UG24AM476
Date:   1776436530

    Initial commit◆


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
62b158ac56ab55aa62b158ac56ab55aa62b158ac56ab55aa62b158ac56ab55aa

--- Object Store ---
Objects created:
8
.pes/objects/05/02010080c060b00502010080c060b00502010080c060b00502010080c060b0
.pes/objects/62/b158ac56ab55aa62b158ac56ab55aa62b158ac56ab55aa62b158ac56ab55aa
.pes/objects/66/b3592c168bc5e266b3592c168bc5e266b3592c168bc5e266b3592c168bc5e2
```

```
thrupthi-g-d@Lily-1: ~/PES1UG24AM476/os-u4-orange-problem
~/PES1UG24AM476/os-u4-orange-problem

commit 62b158ac56ab55aa62b158ac56ab55aa62b158ac56ab55aa62b158ac56ab55aa
Author: PES1UG24AM476
Date:   1776436530

    Add farewelldd farewel!

commit e773b95c2e170b85e773b95c2e170b85e773b95c2e170b85e773b95c2e170b85
Author: PES1UG24AM476
Date:   1776436530

    Update file.txtfarewel

commit ec763b1d8ec763b1ec763b1d8ec763b1ec763b1d8ec763b1ec763b1d8ec763b1
Author: PES1UG24AM476
Date:   1776436530

    Initial commit◆


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
62b158ac56ab55aa62b158ac56ab55aa62b158ac56ab55aa62b158ac56ab55aa

--- Object Store ---
Objects created:
8
.pes/objects/05/02010080c060b00502010080c060b00502010080c060b00502010080c060b0
.pes/objects/62/b158ac56ab55aa62b158ac56ab55aa62b158ac56ab55aa62b158ac56ab55aa
.pes/objects/66/b3592c168bc5e266b3592c168bc5e266b3592c168bc5e266b3592c168bc5e2
.pes/objects/69/b4da6d361b0d8669b4da6d361b0d8669b4da6d361b0d8669b4da6d361b0d86
.pes/objects/ad/56ab55aa552a95ad56ab55aa552a95ad56ab55aa552a95ad56ab55aa552a95
.pes/objects/e7/73b95c2e170b85e773b95c2e170b85e773b95c2e170b85e773b95c2e170b85
.pes/objects/ec/763b1d8ec763b1ec763b1d8ec763b1ec763b1d8ec763b1ec763b1d8ec763b1
.pes/objects/ef/77bbdd6e371b8def77bbdd6e371b8def77bbdd6e371b8def77bbdd6e371b8d

=== All integration tests completed ===
thrupthi-g-d@Lily-1:~/PES1UG24AM476/os-u4-orange-problem$
```

Phase 5 & 6: Analysis-Only Questions

Q5.1 A branch in Git is just a file in .git/refs/heads/ containing a commit hash. Creating a branch is creating a file. Given this, how would you implement pes checkout <branch> — what files need to change in .pes/, and what must happen to the working directory? What makes this operation complex?

- A branch is stored in .pes/refs/heads/<branch> containing a commit hash.
- Steps:
    1. Update .pes/HEAD to point to the branch
    2. Read commit → load its tree
    3. Update working directory to match tree (add/remove/update files)
    4. Update .pes/index accordingly

Complexity:
Requires syncing HEAD, index, and working directory while handling overwrites safely.

Q5.2 When switching branches, the working directory must be updated to match the target branch's tree. If the user has uncommitted changes to a tracked file, and that file differs between branches, checkout must refuse. Describe how you would detect this "dirty working directory" conflict using only the index and the object store.

- Compare working files with index hashes
- If file content ≠ index → modified
- Compare index with target branch tree

Conflict:
If a file is modified AND differs between branches → abort checkout.

Q5.3"Detached HEAD" means HEAD contains a commit hash directly instead of a branch reference. What happens if you make commits in this state? How could a user recover those commits?

- HEAD points to a commit, not a branch
- New commits are created but not linked to any branch

Recovery:
Create a branch pointing to that commit, or use log to retrieve it

Q6.1 Over time, the object store accumulates unreachable objects — blobs, trees, or commits that no branch points to (directly or transitively). Describe an algorithm to find and delete these objects. What data structure would you use to track "reachable" hashes efficiently? For a repository with 100,000 commits and 50 branches, estimate how many objects you'd need to visit.

Algorithm:

1. Start from HEAD and branch refs
2. Traverse commits → trees → blobs
3. Mark reachable objects
4. Delete unmarked objects

Data structure:
Hash set for fast lookup

Estimate:
≈ 100k commits + trees + blobs → hundreds of thousands of objects

Q6.2 Why is it dangerous to run garbage collection concurrently with a commit operation? Describe a race condition where GC could delete an object that a concurrent commit is about to reference. How does Git's real GC avoid this?

Problem:
GC may delete objects that a commit is about to use
Solution (Git):

- Write objects before referencing them
- Use locks and safety checks
- Delay deletion (grace period)